MIL-R-52243B(ME)
12 September 1979
SUPERSEDING
MIL-R-52243A(ME)
7 February 1973

MILITARY SPECIFICATION

RETAINER, BRIDGE PIN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of bridge pin retainer.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
QQ-W-428	- Wire, Steel, High Carbon, (High Carbon Round, for Mechanical Springs, General Purposes).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 5420

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-C-16173	-Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-130	-Identification Marking of US Military Property.
MIL-STD-1188	-Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The bridge retainer shall be as shown on figure 1 and as specified herein.

3.2 First article (preproduction model). The contractor shall furnish one bridge pin retainer for examination within the time frame specified (see 6.2), to prove, prior to starting production, that his production methods will produce bridge pin retainers that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Material shall be as specified herein and on Figure 1. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Copies of analyses and test reports for all materials shall be available for inspection. Chemical composition and mechanical properties of heat-treated materials shall be verified by chemical analysis and mechanical tests (see 6.4).

3.4 Length. The length of the bridge pin retainer shall be as shown on figure 1, as specified (see 6.2).

3.5 Identification marking. The bridge pin retainer shall be identified in accordance with MIL-STD-130.

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3.6 Workmanship

3.6.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. All bends shall be made with controlled means to insure uniformity of size and shape and shall be made to a radius which will not cause cracking at the extreme outer fibers. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not determinetal to safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction bridge pin retainer shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The bridge pin retainer shall be examined as specified herein for the following defects:

MAJOR

101. Nonconformance to the dimensions shown on figure 1.
102. Material not as specified.
103. Heat treatment of bridge pin retainer not as specified.

MINOR

201. Bridge pin retainer not coated as specified.
202. Identification marking incorrect, missing, or not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

104. Containers not as specified for level A or B. Each incorrect container shall be considered one defect.
105. Retainers of unlike size packed together for level A,B, or Commercial.
106. Quantities packed together exceed the limitations specified for level A or B.
107. Strapping not as specified for level A.
108. Marking missing, illegible, incorrect, or incomplete for level A, B, or Commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

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5.1.1 Level A. The retainers of like size shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2) strapping shall be finish A.

5.1.2 Level B. The retainers of like size shall be packed together as specified in 5.1.1 except the box shall be domestic type. As an alternate, the retainers of like size may be packed together in a close-fitting fiberboard box conforming to PPP-B-636, V3c, style CSSC. The gross weight or size of the fiberboard box shall not exceed the limitations of the box specification. Box closure and strapping of either the wood or fiberboard box shall be in accordance with the appendix to the applicable box specification.

5.1.3 Commercial. The retainers of like size shall be packed together in accordance with MIL-STD-1188.

5.2 Marking.

5.2.1 Military. Shipping containers for level A or B shall be marked in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The bridge pin retainers are intended for use in securing bridge pins in fixed and floating military bridges.

6.2 Ordering data. Procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) Length of bridge pin retainer required (see 3.4).
(d) Degree of packing required (see 5.1).
(e) When other than finish B strapping is required (see 5.1.1).

6.3 Preproduction model. Any change or deviations of production bridge pin retainers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish bridge pin retainers conforming to this specification.

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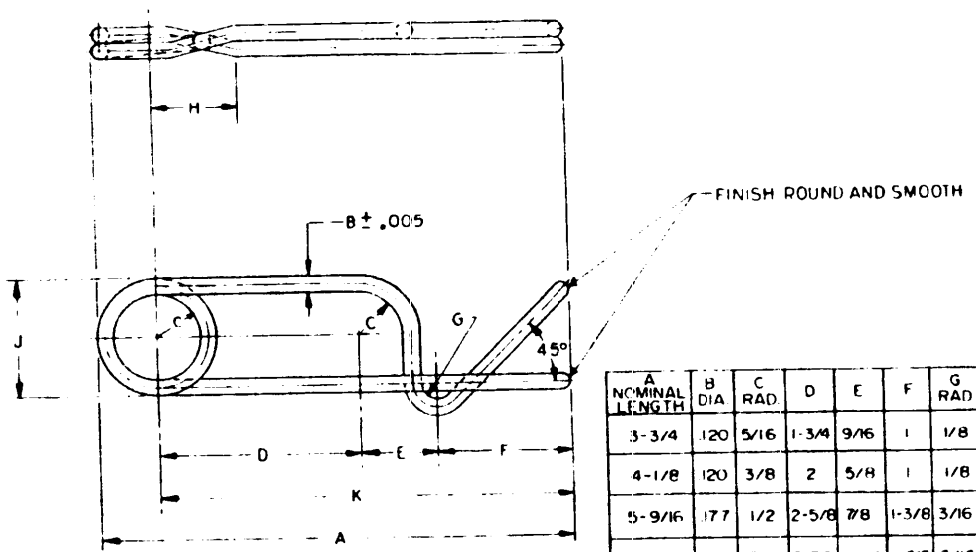
6.4 Recycled Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification. (See 3.3).

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NOTES:

1. SPRING STEEL WIRE SHALL CONFORM TO [illegible] TYPE III, CLASS 1
2. AFTER FORMING INTO SHAPE, THE RETAINER SHALL BE HEAT-TREATED TO PRODUCE A MINIMUM YIELD STRENGTH OF 190,000 PSI AND AN ULTIMATE TENSILE STRENGTH WITHIN THE RANGE OF 200,000 PSI MINIMUM TO 225,000 PSI MAXIMUM
3. AFTER THE HEAT TREATMENT, COAT THE RETAINER WITH A CORROSION PREVENTIVE SOLVENT CUT BACK COATING, HARD FILM TYPE, IN ACCORDANCE WITH MIL-C-[illegible], GRADE 1
4. UNLESS OTHERWISE NOTED, ALL DIMENSIONS ARE IN INCHES, AND TOLERANCES ± 1/16

A NOMINAL LENGTH	B DIA	C RAD	D	E	F	G RAD	H	J APPROX	K	COMMONLY USED WITH CONN PIN OF DIA	FSN
3-3/4	.120	5/16	1-3/4	9/16	1	1/8	3/4	7/8	3 5/16	UP TO 1	5420-060-7030
4-1/8	120	3/8	2	5/8	1	1/8	3/4	1	3 5/8	1 1-1/2	5420-060-7031
5-9/16	177	1/2	2-5/8	7/8	1-3/8	3/16	1	1-3/8	4-7/8	1-1/2 2	5420-060-7032
7-13/16	177	3/4	3-7/8	1-1/8	1-7/8	3/16	1-1/2	1-7/8	6 7/8	2-3	5420-060-7033

FIGURE 1. Retainer, bridge pin

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-52243B(ME) Retainer, Bridge Pin

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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